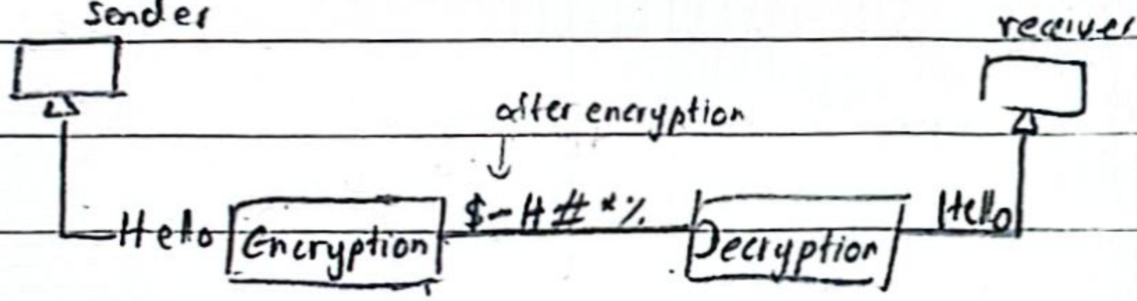
Encryption and Digital Signature

Encryption is the process of converting data into a secret code to prevent unauthorised access.

It's used to protect confidentiality of data during storage or transmission.



There are 2 types of encryptions

1) symmetric encryption - same key for encryp and decryption

2) asymmetric encryption - uses public key to encrypt and private key to decrypt

features of Asymmetric key encryption

Has a ~~public~~ key used publicly
Encryption is done using public key   diagram
Data receiver has to use private key to decrypt data
No need to exchange the key between sender and receiver
Compared with private key, this method is more secure

Are you happy?
↓
Encrypted?
↓
Are you happy + Public key ⟶ a4bc4$y
↓
Decrypt
↓
Private key
↓
Are you happy

Digital Signature

A digital signature is a cryptographic ~~technology~~ technique & used to verify the ~~authentication~~ authenticity, integrity and origin of a digital message or documents.

It looks like a virtual fingerprint confirming the message sent by a specific sender and to certify that it hasn't been altered during the transmission.

How does a digital signature work?

Sender creates a hash (unique digital code) of the message/document.

The hash is encrypted using the senders' private key. (digital signature)

The recipient decrypts the signature using the
sender's public key and it recomputes the
hash of the received message ~~an~~ and fi
it campares both hashes to verify the auth
(sender is verified) and integrity (data wasn't ch

Users of digital signatures

- Signing emails and documents
- Software distribution to prove source and integrity
- e-commerce and online transactions
- Government forms and legal contracts

Network Attacks:

Network attacks are malicious activities targeting computer networks with the intent to steal data, distrupt services or gain unauthorised access.

There are 2 main types of network attacks.

1) Active Attack - An active attack involves the attacker altering, injecting or distupting data or services.

2) Passive Attack - A passive attack occurs when an attacker interrupts or monitors data transmitted without altering it. The goal is to gather information silently.

Common types of network attacks

1) Unauthorized access - The device getting accessed by an unauthorized person without the permission.

2) MITM (Man-in-the-Middle) Attack
Attacker interprets communication between 2 parties and steal log in credentials and etc.

3) DoS (Denial of Service) attack
Overloads a network or service to make it unavailable.

4) Spoofing
Faking a device or user identity to access systems. (IP or email spoofing)

5) Phishing

Fake emails or websites ~~track~~ trick users into giving sensitive information.

6) Port Scan - Unauthorized person entering the system and causing damages by scanning and identifying the ports

make short notes up to 12

7) Eavesdropping - Real time interception of private communication of data or voice by a third party without permission

8) Financial Frauds - Banners to be seen by the internet users mentioning they've won lotteries etc and request insurance charges etc to send them

9) Theft of Proprietary Information

Information being stolen from business establishments by entering the computers through networks

Malware (Malicious Software)

Malware is any software designed to harm, exploit or gain unauthorized access to computer systems, networks or devices.

The following are some effects of virus attacks.
- Speed becomes low
- Generating files without user's acknowledgement
- Decreasing the space of the hard disk
- Taking a lot of time to bootup
- Not opening or dissappearing of files when clicked

Types of malware

Virus - Attaches itself to files or programs and spread when the host ~~is~~ excecutes the program

Worm - These kinds of malware automatically regenerates and spread without the knowledge of the user, often transmitted through the files shared in a network

Ransomware · Encrypts data and demands payment to unlock it.

Spyware - Secretly generates a method to gather user information such as passwords and log in details.

Adware - Displays unwanted ads on different sites.

Spam - Spam is junk or unwanted emails that reach to your email

Trojans - Distinguishes itself as a legitimate software to trick users to install it.

! Hackers

A hacker is someone who's skilled in computers and networking who uses their knowledge to solve problems or explore systems.

There are 3 types of hackers.

White hat hacker - Ethical hackers who test and improve security with permission.

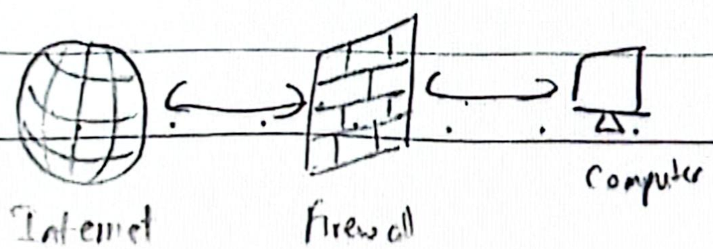
Black hat hackers - They break into information systems, steal or destroy information

Grey hat hackers - Enters an establishment as a whitehat hacker and act ethically wrong and steal information, to do harmful activities.

Security

Firewall

A network security method that can be used to monitor incoming and outgoing network traffic (data) and decides whether to allow or block specific traffic based on a defined set of security rules.



Antivirus programs

Software used to remove virus software and to prevent entering of new viruses

Good habits

Updating Anti-virus programs properly

Defragmenting the hard disk

Avoid downloading software from unreliable websites